



SEMESTER END EXAMINATIONS - JANUARY 2026

Program	: B.E. - Information Science and Engineering	Semester	: V
Course Name	: Frontend Development using React JS	Max. Marks	: 50
Course Code	: 23ISAECS94	Duration	: 2 Hrs

Instructions to the Candidates:

- **PART - A** : Answer all the questions.
- **PART - B** : Answer one full question from each unit.

PART- A

1. React favors Composition over Inheritance. This means that instead of extending classes, UI parts are built by: CO1 (01)
 - a) Combining reusable components
 - b) Using the imperative DOM update style
 - c) Directly manipulating the real DOM
 - d) Writing new <button> elements for every specific case
2. The process of converting JSX code into calls to React.createElement() is known as the: CO1 (01)
 - a) Diffing Algorithm
 - b) Change Detection
 - c) JSX Transform
 - d) Virtual DOM Sync
3. Which lifecycle method is used to call APIs? CO2 (01)
 - a) constructor
 - b) render
 - c) componentDidMount
 - d) componentWillUnmount
4. getSnapshotBeforeUpdate is used in the: CO2 (01)
 - a) Mounting phase
 - b) Updating phase
 - c) Unmounting
 - d) Error handling
5. Inline event handlers are: CO3 (01)
 - a) Recommended for reusable logic
 - b) Good for small actions
 - c) Not allowed
 - d) Replace addEventListener
6. Which of the following is a controlled component? CO3 (01)
 - a) <input defaultValue="abc" />
 - b) <input value={x} onChange={f} />
 - c) <input type="file" />
 - d) <textarea></textarea>

7. useParams returns: CO4 (01)
a) URL parameters
b) Query string
c) Location object
d) Component props
8. Link component prevents: CO4 (01)
a) JSX errors b) Page reload c) DOM mutation d) API calls
9. Environment variables in React must start with: CO5 (01)
a) VAR_ b) ENV_ c) REACT_APP_ d) APP_ENV_
10. Which step generates optimized JavaScript/CSS files? CO5 (01)
a) Development mode
b) Production build
c) Testing mode
d) Node server

PART – B UNIT – I

1. a) Explain the role of ReactDOM in connecting React components with the browser DOM. CO1 (04)
b) Discuss declarative and imperative programming with examples. CO1 (04)
2. a) Explain how JSX is compiled internally with Babel transformations. CO1 (04)
b) Discuss JSX syntax rules with suitable examples. CO1 (04)

UNIT – II

3. a) Write a functional component that accepts props for name and marks and displays grade. CO2 (04)
b) Write differences between props.children and normal props with an example. CO2 (04)
4. a) Write a Parent and Child component where parent passes a message through props. CO2 (04)
b) Explain how state flows inside a React application with an example. CO2 (04)

UNIT – III

5. a) Describe event bubbling in React with an example. CO3 (04)
b) Build a functional component where users can type into an input box, and the app displays the current character count below it. CO3 (04)
6. a) Discuss handling inputs like checkbox, and textarea with examples. CO3 (04)
b) Why is lifting state up required? Provide real-time example. CO3 (04)

UNIT – IV

7. a) Discuss performance optimization using React DevTools Profiler. CO4 (04)
b) Create a multi-page app with Home, Services, Contact pages using Routes. CO4 (04)
8. a) Create a navigation bar that highlights the active page using NavLink. CO4 (04)
b) Explain how dynamic routing works using :id parameters. CO4 (04)

UNIT – V

9.
 - a) Explain the purpose of .env file and how environment variables are used in React. CO5 (04)
 - b) Create a card layout where top-level styles are global but component styles use CSS Modules. CO5 (04)
10.
 - a) Explain useState with an example of updating state objects/arrays. CO5 (04)
 - b) Create a component that shows current time and updates every second using useEffect. CO5 (04)
